

TCS-693

B. TECH. (CSE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

FULL STACK WEB DEVELOPMENT

Time : 1½ Hours

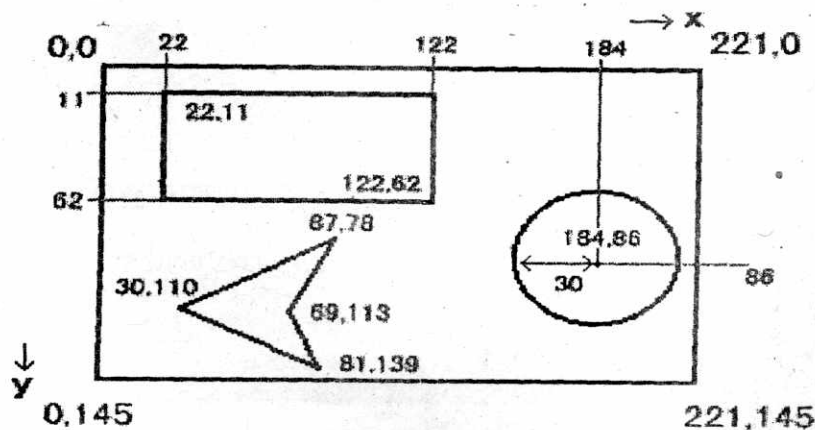
Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Why are metatags used in HTML ? Where we use metatags ? Explain the essential key attributes of a metatag, with suitable examples. Also explain tags used in <head> of html. (CO1, CO2)

OR

- (b) Explain image map with all required attributes area and shapes, Image map usability in HTML. Write the code in HTML for this image map : (CO1, CO2)



P. T. O.

2. (a) What is the Box model in CSS ? Which CSS properties are a part of it ?
What are the advantages of using CSS ? What are the limitations of CSS ?

(CO1, CO2)

OR

- (b) How to include CSS in the webpage ? What are the different types of Selectors in CSS ? What are Pseudo elements and Pseudo classes ?
3. (a) Write code in HTML to implement this Form.

(CO3)

XYZ College/School
Student Registration Form

Student Image: (less than 5 Mb)

Student Name:

Father's Name:

Mother's Name:

Gender: ☐ Male ☐ Female ☐ Other

Date of Birth:

E-mail:

Level:

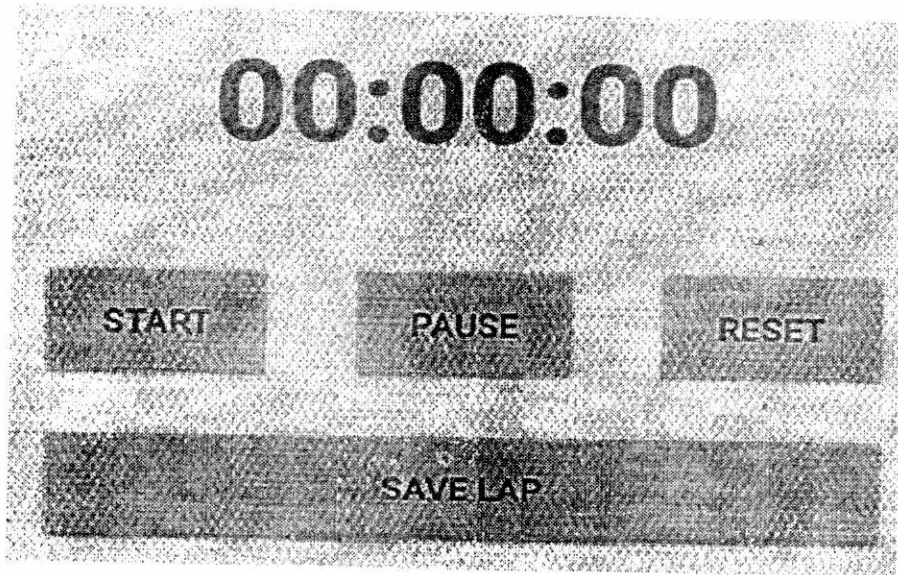
Department:

Tel/Mobile:

(3)

OR

- (b) What exactly do you know about JavaScript ? Write code using HTML, CSS and JavaScript to implement stopwatch : (CO3)



4. (a) What is jQuery ? How is jQuery different from other JavaScript frameworks ? What is the use of CSS() method in jQuery ? Implement CSS() method in JQuery with suitable example. (CO3)

OR

- (b) What is Ajax ? What are Ajax applications ? How Ajax objects can be created ? What are the protocols, status and ready State used by Ajax ? (CO3)
5. (a) How is XML different from HTML ? What is a well formed XML document ? What is Document Type Definition (DTD) ? Explain with suitable examples. (CO2)

OR

- (b) Write a JavaScript program that accepts a month in the format mm/yyyy and display the month's calendar : (CO2)

Output :

Test Data : ('1/2019') →

"	M	T	W	Th	F	S	Su	"
"		1	2	3	4	5	6	"
"	7	8	9	10	11	12	13	"
"	14	15	16	17	18	19	20	"
"	21	22	23	24	25	26	27	"
"	28	29	30	31	"			

('2/1990') →

"	M	T	W	Th	F	S	Su	"
"			1	2	3	4	5	"
"	6	7	8	9	10	11	12	"
"	13	14	15	16	17	18	19	"
"	20	21	22	23	24	25	26	"
"	27	28	"					

Roll No.

TCS-604

**B. TECH. (CSE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023**

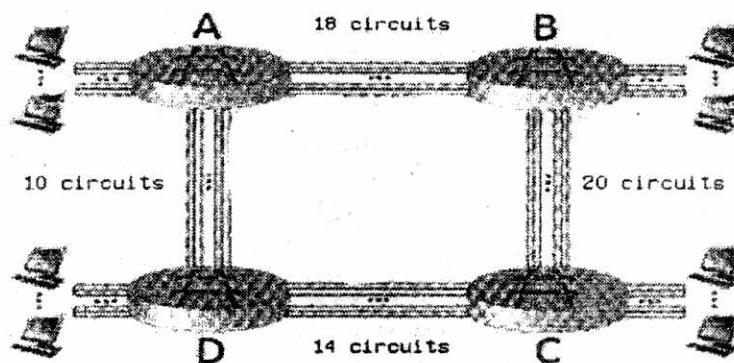
COMPUTER NETWORKS-I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain how packet switching is different from circuit switching. Consider the circuit-switched network shown in the figure below, with circuit switches A, B, C, and D. Suppose there are 18 circuits between A and B, 20 circuits between B and C, 14 circuits between C and D, and 10 circuits between D and A. Suppose that every connection requires 2 consecutive hops, and calls are connected clockwise. For example, a connection can go from A to C, from B to D, from C to A, and from D to B. With these constraints, what is the maximum number of connections that can be ongoing in the network at any one time ? (CO1)



P. T. O.

- (b) Discuss the TCP/IP protocol suite on the basis of protocol layering principle. (CO3)
2. (a) Calculate the latency (from first bit sent to the last bit received) for the following : (CO1)
- (i) 1 Gbps Ethernet with a single store and forward switch in the path and a packet size of 2500 bits. Assume that each link introduces a propagation delay of 20 microsecond and that the switch begins retransmitting immediately after it has finished receiving the packet.
- (ii) Same as (i) but with three routers.

OR

- (b) Consider a message that is 8×10^6 bits long that is to be sent from Host A to Host B with two packet-switches in between. Suppose each link is 2 Mbps. Ignore propagation, queuing, and processing delays : (CO1)
- (i) Consider sending the message from A to B without message segmentation. How long does it take to move the message from Host A to the first packet switch ? What is the total time to move the message from A to B ?
- (ii) Now suppose that the message is segmented into 800 packets, with each packet being 10,000 bits long. How long does it take to move the first packet from A to the first switch ? When the first packet is

being sent from the first switch to the second switch, the second packet is being sent from A to the first switch. At what time will the second packet be fully received at the first switch ? (CO1)

3. (a) Assume that a client wants to retrieve `www.cnn.com` but has no information about the webserver's IP address. Describe the process of the client obtaining the IP address under the assumption that it is not cached at the local DNS server and the local DNS server not cached an entry for the .com DNS server. (CO2)

OR

- (b) Differentiate between the following : (CO1)

(i) POP and IMAP

(ii) Transmission Delay and Propagation Delay

4. (a) Describe the components of HTTP protocol. Suppose you needed to use HTTP to download a web page with three embedded images. Draw diagrams, similar to those from class, depicting the main interactions between the client and server when using non-persistent HTTP, persistent HTTP without pipelining, and persistent HTTP with pipelining. (CO2)

OR

- (b) Discuss the working of simple mail transfer protocol with suitable example. How does MIME enhance SMTP ? (CO2)

P. T. O.

5. (a) Consider distributing a file of $F = 25$ Gbits to N peers. The server has an upload rate of $u_s = 30$ Mbps, and each peer has a download rate of $d_i = 2$ Mbps and an upload rate of u . For $N = 10, 100$, and $u = 300$ Kbps, 700 Kbps, prepare a chart giving the minimum distribution time for each of the combination of N and u for both client-server distribution and P2P distribution. (CO2)

OR

- (b) Explain the complete process used in p2p communication using Bit Torrent. (CO2)

Roll No.

TCS-611

**B. TECH. (CS/CC/BA) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023**

SOFTWARE ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Discuss Agile manifesto and principles. Also how agile principles are followed in eXtreme Programming (XP) ? (CO1)

OR

- (b) Enlist the characteristics of software ? Explain with suitable example. (CO1)

2. (a) ABC is an international software house. ABC is currently working on a project that is totally new for the development team and even the client is confused about the requirements of this project. Hence this company is facing difficulties because they fail to apprehend user requirements properly. For this project, it is decided to build a sample application and show it the client for feedback. In the context of this above scenario as a project manager what will be the choice of the software lifecycle model ? Explain with suitable reason. Elaborate the advantages and disadvantages of the identified model. (CO2)

P. T. O.

(2)

OR

- (b) List any *two* techniques used for elicitation requirements. Compare the two techniques and list where each is applicable. (CO2)

3. (a) What is data dictionary ? How is it used in software engineering ? (CO2)

OR

- (b) Develop the software requirements document for the following requirements : A coffee vending machine serves coffee to customers. A customer can choose a type of coffee among a list of options, supply the amount required and get served. Each coffee is prepared by adding units of hot water, coffee powder, milk and sugar. The recipe for each coffee is stored. (CO2)

4. (a) What minimum features are required to be present in a good SRS ? (CO1)

OR

- (b) Draw a DFD for the following : (CO1)
- (i) money withdraw from ATM.
 - (ii) Safe home Software.

5. (a) What are the characteristics of a good design ? Describe different types of coupling and cohesion. How design evaluation is performed ? (CO1)

OR

- (b) Explain how software is developed using Rapid Application Development (RAD) model. Also elaborate the key features that leads to rapidity. (CO2)

Roll No.

TCS-601

B. TECH. (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
(All Branches)
COMPILER DESIGN
Time : 1½ Hours
Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain the C compilation process with the help of a block diagram. (CO1)

OR

- (b) Write the LL(1) parsing algorithm. Check whether the following grammar is LL(1). If not transform it into LL(1) if possible, without changing grammar semantics : (CO1)

$S \rightarrow aA \mid AB, A \rightarrow Ab \mid c, B \rightarrow e \mid f.$

2. (a) What is full bootstrapping ? We have a compiler written in C which translates java to assembly. Create another compiler that is written in C++ and translates Java to assembly. (CO1)

OR

- (b) Compute first and follow functions for the following grammar : (CO1)

$G = (\{S, A, B, C\}, \{a, b, g, h\}, P, S).$

$S \rightarrow ACB \mid CbB \mid Ba$

$A \rightarrow da \mid BC$

$B \rightarrow g \mid \epsilon$

$C \rightarrow h \mid \epsilon$

P. T. O.

3. (a) Show the grammar is not SLR but CLR :

(CO2)

$$S \rightarrow L = R \mid R$$
$$R \rightarrow L$$
$$L \rightarrow * R \mid id$$

OR

- (b) Check whether the following grammar is LR (0) or not :

(CO2)

$$S \rightarrow dA \mid aB$$
$$A \rightarrow bA \mid c$$
$$B \rightarrow bB \mid c$$

4. (a) Construct an operator precedence table for the given grammar and construct the operator precedence function table from the operator relation table :

(CO2)

$$A \rightarrow B * C / C$$
$$C \rightarrow B$$
$$B \rightarrow + C / id$$

OR

- (b) Write short notes on the following :

(CO2)

(i) Run-Time Semantic vs. Compile Time Semantic

(ii) Compiler vs. Interpreter

(iii) Shift-reduce vs. Reduce-reduce conflict.

(iv) High-level language vs. Pure high-level language

(3)

5. (a) Design an NFA over the input 0,1 which accepts the string having the third last symbol 0 from the right. Also, convert it into an equivalent DFA. (CO2)

OR

- (b) Describe major issues with top-down parsing. Write a recursive descent parser procedure for the following grammar : (CO2)

$$A \rightarrow aB, B \rightarrow +aB \mid \epsilon$$

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TCS-671

B. TECH. (CSE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

BIGDATA STORAGE AND PROCESSING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Flipkart is a privately hosted e-commerce platform specializing in online retail, fashion e-commerce, mobile phones, etc. Analyze a scenario where Flipkart wants to benefit from big data. Develop a plan for capturing and organizing this data to support their business objectives. (CO1)

OR

- (b) Dr. Lal PathLabs Limited is an Indian diagnostic and related healthcare testing provider. Evaluate the challenges of managing and analyzing big data in the healthcare sector. Propose a solution that addresses these challenges and discuss the potential benefits to the healthcare sector.

(CO1)

P. T. O.

2. (a) Punjab National Bank (PNB) provides various financial and banking services, including Credit/Debit Cards, Savings/Current Accounts, Loans, Investments, etc. Describe the technological challenges associated with big data in the banking sector. Develop a plan to overcome these challenges and discuss the potential benefits to the banking industry. (CO1)

OR

- (b) Micro Focus offers multiple enterprise software products, solutions, and services for businesses looking to modernize core data and IT operations. Micro Focus has recently adopted Hadoop Distributed File System. Analyze the design of HDFS and discuss how it differs from traditional file systems. (CO1)

3. (a) Zee Entertainment Enterprises Limited (ZEEL) is a media and entertainment company providing broadcasting services. The company operates through content and broadcasting segment. ZEEL wants to analyze customer data to improve advertising strategies. The company has implemented HDFS in its organization. Discuss the roles of the DataNode and NameNode, in the HDFS architecture, and evaluate their effectiveness in managing and storing big data. (CO2)

OR

- (b) A government organization has implemented HDFS in their organization. Discuss the limitations of HDFS and propose solution to overcome these limitations. (CO2)

(3)

4. (a) iTechArt has been the partner of choice for fast-growing startups and innovative companies since 2002, providing fully dedicated engineering teams and custom software solutions. iTechArt is experiencing issues with data integrity in their Hadoop I/O. Analyze the concept of data integrity in Hadoop I/O, and discuss methods to ensure data integrity.

(CO2)

OR

- (b) L & T Finance Limited was established in 1994 and is headquartered in Mumbai. It offers funding services to agriculture, trade, and industry sectors. Evaluate the features, types, and formats of the MapReduce framework. Develop a plan to use the framework to process and analyze large datasets for L & T Finance Limited.

(CO2)

5. (a) JD.com is an e-commerce retailer that sells electronics products and general merchandise products, including audio, video products, and books. JD.com retail company has implemented Hadoop Distributed File System and Hadoop I/O in their organization. Analyze their effectiveness in capturing, organizing, and analyzing big data to support their business objectives.

(CO2)

OR.

- (b) Fortis Healthcare Limited (FHL) is an Indian multinational chain of private hospitals headquartered in India. Fortis started its health care operations in Mohali, Punjab, where the first Fortis hospital was started. FHL has implemented Hadoop Distributed File System and Hadoop I/O in their organization. Analyse their effectiveness in managing and processing big data to support their objectives.

(CO2)

Roll No.

TCS-691

B. TECH. (CSE, AI & DS) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

IMAGE PROCESSING AND COMPUTER VISION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Why the fundamental steps of digital image processing are required in image processing and computer vision ? Explain with the help of any real-time image processing application. (CO1)

OR

- (b) Consider two gray images have the same histogram. Write your comments on the following and justify your answer with suitable example : (CO1)

- (i) Is the resolution of both images equal or not ?
- (ii) Are the contrast and brightness of both images same or different ?
- (iii) Is the total storage size equal or not ?
- (iv) Are both input images same ?

P. T. O.

2. (a) Why bit plane slicing is required for digital image processing ? How 8-bit gray-scale image can be converted into an equivalent of the 2-bit gray-scale image ? Explain and justify your answer with a suitable example. (CO1)

OR

- (b) How much storage capacity is required to store image with size of 1024×768 and 256 gray levels ? Also identify the number of bits required to store the pixel value. (CO1)
3. (a) Why edge detection is most common approach for detecting discontinuities ? Consider a 5×5 image matrix and apply 3×3 Sobel operator to detect edges on works for detecting discontinuities in images. (CO2)

OR

- (b) Suppose that a 3-bit image ($L = 8$) of size 5×5 pixels as shown in below matrix, where the intensity levels are integers in the range $[0, L-1]$: (CO2)

1	1	0	4	0
0	0	0	1	2
1	2	1	1	3
4	0	2	0	3
0	4	0	1	0

- (i) Compute the histogram for given matrix.
(ii) Calculate transformation function of histogram.
(iii) Compute the equalized histogram.

(3)

4. (a) How smoothing filter works in image processing ? Which method is effectively works for image smoothness in spatial domain ? Apply that method by considering a 5×5 image matrix for enhancing smoothness in the images. (CO2)

OR

- (b) Why sharpening filters are required in image processing ? Which method is effectively works for image sharpness in spatial domain ? Apply that method by considering a 5×5 image matrix for enhancing sharpness in the images. (CO2)
5. (a) Why does the noise considered to be an additive in images ? How does median filter work on the noisy image ? Explain each and every step by apply 3×3 median filter on below Image Matrix : (CO3)

$$\begin{bmatrix} 6 & 5 & 4 & 3 & 2 & 12 \\ 5 & 4 & 3 & 2 & 12 & 2 \\ 4 & 3 & 2 & 12 & 2 & 3 \\ 3 & 2 & 12 & 2 & 3 & 4 \\ 2 & 12 & 2 & 3 & 4 & 5 \\ 12 & 2 & 3 & 4 & 5 & 6 \end{bmatrix}$$

OR

- (b) Find with in class variance using Otsu's thresholding for below matrix by considering a single threshold value : (CO3)

$$\begin{bmatrix} 0 & 1 & 1 & 2 \\ 2 & 5 & 5 & 4 \\ 6 & 7 & 6 & 2 \\ 4 & 6 & 5 & 0 \end{bmatrix}$$

Roll No.

TCS-651

B. TECH. (CSE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

DevOps ON CLOUD

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Which are the top DevOps tools ? How do all these tools work together ? (CO1, CO3)

OR

- (b) Describe the use of Source code management system. Explain any *five* "git" commands. (CO1, CO3)

2. (a) How can we schedule a job in Jenkins ? Write down the steps for configure Poll SCM in Jenkins. (CO1, CO2, CO3)

OR

- (B) Explain Audit Trail and the advantages of using Audit Trail plugin in Jenkins. (CO1, CO2, CO3)

3. (a) Explain the evolution of software development. (CO2, CO3)

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(2)

OR

(b) What is Git rebase and how can it be used to resolve conflicts in a feature branch before merge ? (CO2, CO3)

4. (a) Write down any *four* plugins you have used with Jenkins and their usage. (CO1, CO3)

OR

(b) What is Jenkins pipeline ? Explain some common practices of CI/CD.

(CO1, CO3)

5. (a) Summarize Selenium and its components. (CO2, CO3)

OR

(b) Write down the advantages and disadvantages of using selenium, as well as write a code (in any programming language) which can open the link '<https://www.google.com/>' and type 'DevOps' in the search field.

(CO2, CO3)

Roll No.

TCS-619

B. TECH. (CSE/CE-SPL) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

NETWORK AND SYSTEM SECURITY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Design a security protocol for the secure session key establishment between two parties, i.e., party A and party B. You have to use the mechanism of public key certificates of public key distribution. (CO4)

OR

- (b) Use Hill cipher scheme to compute the plaintext of "DNFWGH", where key is "NOTONEOUT" and $n = 26$. (CO4)

2. (a) Design a RSA algorithm-based secure communication system in which you have to provide protection for data leakage attack and replay attack. Further, achieve the authenticity of the exchanged messages using the RSA algorithm. (CO6)

P. T. O.

(2)

OR

(b) How would you compare chosen plaintext attack and chosen ciphertext attack ? A cryptographic system vulnerable to ciphertext only attack is completely insecure. Justify this statement. (CO6)

3. (a) In a RSA algorithm-based cryptographic system, the values of different parameters are $p = 13$, $q = 5$ and $e = 7$. What is the value of the private key (d, n) . Further obtained the plaintext of "46" using the decryption procedure of RSA algorithm ? (CO3)

OR

(b) State and prove that double Data Encryption Standard (DES) algorithm is vulnerable to meet-in-the-middle attack. (CO3)

4. (a) How would you compare unconditionally secure and computationally secure cryptographic schemes ? (CO2)

OR

(b) Describe the following general security requirements : (CO2)
Non-repudiation, Authentication and Forward secrecy.

5. (a) Explain the encryption and decryption procedures of the Vernam cipher. How is the Vernam cipher scheme perfectly secure ? (CO1)

OR

(b) Describe the model for network security using a figure, where this model is applicable. (CO1)

Roll No.

TCS-622

B. TECH. (CSE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
CLOUD COMPUTING TECHNOLOGIES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) "Virtualization is a major enabler for cloud computing." Elaborate on the statement and discuss the role of virtualization in cloud computing. (CO3)

OR

- (b) How do you classify different types of cloud based services ? (CO1)
2. (a) What are the advantages and disadvantages of using cloud computing as compared to the traditional means of using computation services ? (CO1)

OR

- (b) Write short notes on any *two* of the following : (CO2)
- (i) Grid computing
 - (ii) Parallel computing
 - (iii) Distributed computing

P. T. O.

(2)

3. (a) Present at least *five* different cloud based services for any *one* of the following cloud service providers : (CO2)

- (i) Google Cloud Platform
- (ii) Microsoft Azure
- (iii) Amazon Web Services

OR

- (b) Write short notes on any *two* of the following : (CO1)

- (i) Edge computing
- (ii) Fog computing
- (iii) Utility computing

4. (a) Write a short note on evolution of Web, tracing the journey from Web 1.0 to Web 2.0 to Web 3.0. (CO2)

OR

- (b) Discuss the total cost of ownership for cloud computing. How would you categorize various types of cost involved in traditional processing ? (CO3)

5. (a) Discuss the cloud service delivery models and cloud service deployment models along with their advantages and disadvantages. (CO1)

OR

- (b) Discuss the vision of cloud computing. According to you, what are the milestones for achieving this vision ? (CO2)

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TCS-624

B. TECH. (CSE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

SECURITY IN SOCIAL NETWORK

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) List some of the features of Python. Give the rules for naming an identifier in Python. Draw and explain multi-way if-elif-else statements.

(CO1)

OR

- (b) Define List and which type of operations perform on list. Write a python program that counts odd and even numbers from a list. (CO1)

2. (a) Difference between static and dynamic web pages. What is a form ? What are the major attributes of the form ? Explain any *six* form components with example.

(CO1)

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(2)

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OR

- (b) Write a program in HTML to implement the given table : (CO1)

Day	Seminar		
	Schedule		Topic
	Begin	End	
Monday	8:00 a.m.	5:00 p.m.	Introduction to XML
			Validity : DTD and Relax NG
Tuesday	8:00 a.m.	11:00 a.m.	XPath
	11:00 a.m.	2:00 p.m.	
	2:00 p.m.	5:00 p.m.	XSL Transformations
Wednesday	8:00 a.m.	12:00 p.m.	XSL Formatting Objects

3. (a) What are pros and cons of social media ? Explain with the help of an example. (CO2)

OR

- (b) Discuss the Architecture of Web and working with social media APIs. (CO3)

4. (a) How do I get tweets from Twitter API in Python ? (CO2)

OR

- (b) What are the privacy and security concerns in web and social media platforms ? (CO2)

(3)

5. (a) What are the steps involved in data pre-processing text for Natural Language Processing ? (CO2)

OR

- (b) Write short notes on the following : (CO2)
- (i) Cross-Site Scripting (XSS)
 - (ii) Denial of Service (DoS)
 - (iii) Client-Server Architecture

Roll No.

TCS-642

B. TECH. (CSE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

BIG DATA ANALYTICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) ROI MINDS is a full-scale digital marketing agency that provides performance and growth marketing services to brands looking for hyper-accelerated growth. Imagine you are a data analyst at ROT MINDS, a social media company. Describe how you would analyze real-time social media streams to identify trending topics and user sentiment. (CO1)

OR

- (b) Trip.com is China's leading provider of travel services, including reservations for accommodation, transportation, tours, and corporate travel. Analyze the challenges of Trip. com, an online travel agency, faces in analyzing the vast amounts of data generated by customer

P. T. O.

bookings, cancellations, and preferences and provides recommendations to improve the analysis process. (CO1)

2. (a) YouTube Music is a streaming service with official singles, remixes, live performances, and videos. You can use this application for iOS, Android, and desktop. YouTube Music wants to analyze user data to improve recommendations and user experience. Design a data model and architecture that can handle large volumes of streaming data. (CO1)

OR

- (b) Consider a real-life scenario from the healthcare industry. Describe the process of collecting and analyzing data to identify potential health risks and improve patient outcomes. (CO1)

3. (a) Analyze the importance of descriptive analytics for a retail business that wants to gain insights into customer behavior and preferences. Provide examples of how descriptive analytics can be used to improve marketing and sales strategies. (CO2)

OR

- (b) Sun TV Network Limited is engaged in providing broadcasting services. The company operates through the media and entertainment segment. Sun TV Network Limited wants to analyze customer data to improve advertising strategies. Design a data collection process that can capture relevant data from various sources and explain the steps involved in analyzing the data. (CO2)